

Awais Fateh Ali

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SUMMARY

Results-driven Software Engineer with hands-on experience in MERN stack development and introductory knowledge of Machine Learning. Skilled in building web applications, designing RESTful APIs, and integrating AI/ML models into full-stack systems. Proven ability to deliver end-to-end solutions with clean architecture and efficient database design.

PROFESSIONAL EXPERIENCE

Knowledge Stream (Samsung Innovation Campus)

Lahore

AI Trainee

3 Months

- Built and trained machine learning models using Python, improving prediction accuracy through data preprocessing and feature engineering.
- Implemented core ML concepts including supervised learning, model evaluation, and optimization techniques.
- Worked with datasets to analyze patterns and improve model performance by 20%+.
- Strengthened debugging and problem-solving skills by handling real-world AI workflows.

PROJECTS

DARUT | *Roman Urdu Depression Detection* | MERN Stack + FastAPI | Live

- Developed a full-stack Roman Urdu mental health chat application using React, Redux, Node.js, and FastAPI, enabling real-time depression pattern detection across a three-service distributed architecture.
- Trained a TF-IDF and Logistic Regression NLP model served via FastAPI, integrated with Gemini API to generate confidence scores and personalized mental health advice based on user chat history.
- Engineered a five-message analysis pipeline with color-coded depression severity badges, delivering structured insights with measurable confidence scores to support early mental health intervention.

Multi-Vendor E-Commerce Platform | *MERN Stack*

- Built a scalable multi-vendor e-commerce platform using the MERN stack, supporting multiple sellers with independent product management, order processing, and authentication workflows.
- Designed optimized MongoDB schemas and RESTful APIs to handle high-volume transactions, improving application performance by 30% through efficient state management and API optimization.
- Implemented secure JWT authentication, cart system, and payment workflow simulation, enabling end-to-end purchase flow across buyer and seller roles.

Digital Agency | *Agency Website + Admin Panel* | MERN Stack | Live

- Developed a full-stack digital agency platform using MongoDB, Express, React, and Node.js, featuring a client-facing service showcase with contact form and a secure JWT-authenticated admin panel.
- Built full CRUD functionality for service management using RESTful APIs with Express.js and MongoDB, enabling admins to dynamically add, update, and remove offerings without code changes.
- Designed a responsive admin dashboard with role-based access control, centralizing contact submission tracking and service management to streamline agency operations.

QuickCart | *E-Commerce Website* | Next.js | Live

- Built a full-stack e-commerce platform using Next.js with server-side rendering, supporting product browsing, order management, and a responsive storefront UI across all devices.
- Integrated Cloudinary for media storage and optimized product image delivery, reducing load times and improving visual performance across the storefront.
- Implemented secure user authentication and session management using Clerk, enabling streamlined onboarding and protected order management workflows.

Gradio-AI | *Assignment Grading System* | MERN Stack | Live

- Developed a full-stack assignment grading platform using MongoDB, Express, React, and Node.js, enabling teachers to create, manage, and grade assignments through a structured classroom workflow.
- Implemented JWT-based authentication and RESTful APIs using Express.js, securing teacher sessions and enabling seamless frontend-backend communication across admin and teacher roles.
- Reduced manual grading effort by 40% by automating assignment submission and grading workflows, with a responsive React UI built using Tailwind CSS for cross-device accessibility.

SKILLS

Languages: Python, C++, JavaScript, PHP

Frameworks: MERN Stack (MongoDB, Express.js, React.js, Node.js), FastAPI, Next.js

Databases: MongoDB, MySQL, PostgreSQL

Tools: Git, GitHub, VS Code, Postman, Google Colab

AI/ML: Scikit-learn, TensorFlow, Pandas, NumPy, TF-IDF, Logistic Regression, XGBoost

Concepts: Data Structures & Algorithms, REST APIs, Software Development Lifecycle, OOP, NLP

ACHIEVEMENTS

Published Research Paper – *RAM-BASED CACHING MODEL IMPROVING PERFORMANCE OF DISTRIBUTED SYSTEMS*, International Journal of Social Sciences Bulletin (IJSSB), 2025. Published peer-reviewed research on cache optimization in distributed systems, focusing on performance improvement techniques in memory management. [Link](#)

Solved **150+** **problems** on LeetCode, strengthening data structures and algorithmic problem-solving.

EDUCATION

University of Education

BS Computer Science | CGPA: 3.68

Lahore

2022 – 2026

- **Relevant Coursework:** Data Structures, Algorithms, Operating Systems, Databases, Computer Networks